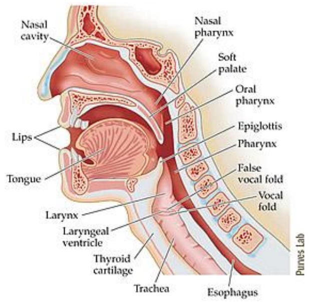
# Phonology

Phonemes and allophones

# Class 4

# Phonetics vs. Phonology

- Phonetics
  - Provides us with an **inventory** and **description** of all possible sounds found in human speech
    - How they're produced (**articulatory features**: voicing, place, manner, nasality…)
    - What their **physical properties** are




  - Doesn't tell us:
    - Which sounds are *meaningful* in a given language
    - How the sounds in a given language can (and can't) be combined

# Phonetics vs. Phonology

- Phonology investigates…
  - how sounds are **distributed** and **organized** into **patterns** in different languages
  - which differences count as **meaningful** differences
  - how sounds **interact systematically** with other sounds in the course of normal speech
  - the **cognitive** organization of sounds in a speaker’s mind (part of linguistic competence)
  - “Language isn’t just in the mouth or ears, but also in the brain” (p. 38)

# Psychological vs. physical reality

/t/

| *top* | *stop* | *cat* | *little* | *kitten* | *center* |
|---|---|---|---|---|---|
| [tʰ ] | [t] | [t̚ ] | [ɾ] | [ʔ] | [ ] |

**Psychological reality**: the idea of “t”
(for native speakers of English)

**Physical reality**: they are pronounced differently.
(acoustic phonetic fact)

# Phonemes vs. allophones

/t/

Phoneme

(psychological reality)

| *top* | *stop* | *cat* | *little* | *kitten* | *center* |
|---|---|---|---|---|---|
| [tʰ ] | [t] | [t̚ ] | [ɾ] | [ʔ] | [ ] |

***Allophones***

(actual pronunciations)

- **/Phoneme/**
  - A class of speech sounds that are judged by a native speaker to count as **the 'same' sound**
- **[Allophone]**
  - Physical realization of a phoneme

# Types of distribution

- How can you tell if sounds count as allophones of the same phoneme vs. allophones of different phonemes in a given language?

→ Look at the distribution

(Ask: What is the set of phonetic environments in which particular sounds occur?)

- Types of distribution
  - Contrastive distribution
  - Complementary distribution
  - Free variation

**Important concept:**

**MINIMAL PAIRS**

# Minimal pair

- A pair of words which
  - are pronounced in the same way except for <u>one sound that differs</u>

  AND correspondingly

  - have <u>different meanings</u>

# Minimal pairs (in 'standard American' English)

| | | | |
|---|---|---|---|
| /pɪt/ | ‘pit’ | /bɪt/ | ‘bit’ |
| /næpɪŋ/ | ‘napping’ | /næbɪŋ/ | ‘nabbing’ |
| /kæp/ | ‘cap’ | /kæb/ | ‘cab’ |
| | | | |
| /deɪ/ | ‘day’ | /ðeɪ/ | ‘they’ |
| /hɛdəɹ/ | ‘header’ | /hɛðəɹ/ | ‘Heather’ |
| /leɪd/ | ‘laid’ | /leɪð/ | ‘lathe’ |

# Contrastive distribution

- In English, the words

  *day* /**d**eɪ/ and *they* /**ð**eɪ/

  form a **minimal pair**.

  - They differ in only a single sound: /**d**/ vs. /**ð**/
  - They have different meanings
  - In English, these sounds are **contrastive**

- What about [**d**] vs. [**ð**] in Spanish?

# [d] vs. [ð] in Spanish

| | | | |
|---|---|---|---|
| [**d**ama] | ‘lady’ | [la**ð**o] | ‘side’ |
| [**d**emonias] | ‘demons’ | [universi**ð**a**ð**] | ‘university’ |
| [**d**ulse] | ‘sweet’ | [imbali**ð**o] | ‘invalid’ |
| [**d**isfras] | ‘disguise’ | [grena**ð**a] | ‘Grenada’ |

- Is the distribution of [d] and [**ð**] predictable?
  - /d/ at the beginning of words → [**d**]
  - /d/ in other envionrments → [**ð**]

# Complementary distribution

- In Spanish, [d] and [ð] are in **complementary distribution**.
  - You can **predict** which one will appear in a given context
  - The sounds are **non-contrastive** in Spanish
  - They are **allophones** of the same phoneme: /d/

# Visualization

| English | Spanish |
| --- | --- |
| /d/ /ð/ | /d/ |
| [d] [ð] | [d] [ð] |

**English**

Contrastive
Unconditioned
Unpredictable

Allophones of ***different*** phonemes

**Spanish**

Complementary
Conditioned
Predictable

Allophones of the ***same*** underlying phoneme

# Visualization (an analogy)

/Shin'ichi/ /Ai/ /Shin'ichi/

[Shin'ichi] [Ai] [Shin'ichi] [Konan]

**English**

Contrastive
Unconditioned
Unpredictable

Allophones of ***different*** phonemes

**Spanish**

Complementary
Conditioned
Predictable

Allophones of the ***same*** underlying phoneme

# Summary in prose

- In Spanish, [d] and [ð] are…
  - in **complementary distribution** – appearing in different phonetic environments ([d] word-initially vs. [ð] between vowels)
  - **conditioned** – occurring for explainable articulatory reasons
  - **predictable** – you know which version will be produced based on where it appears in a word

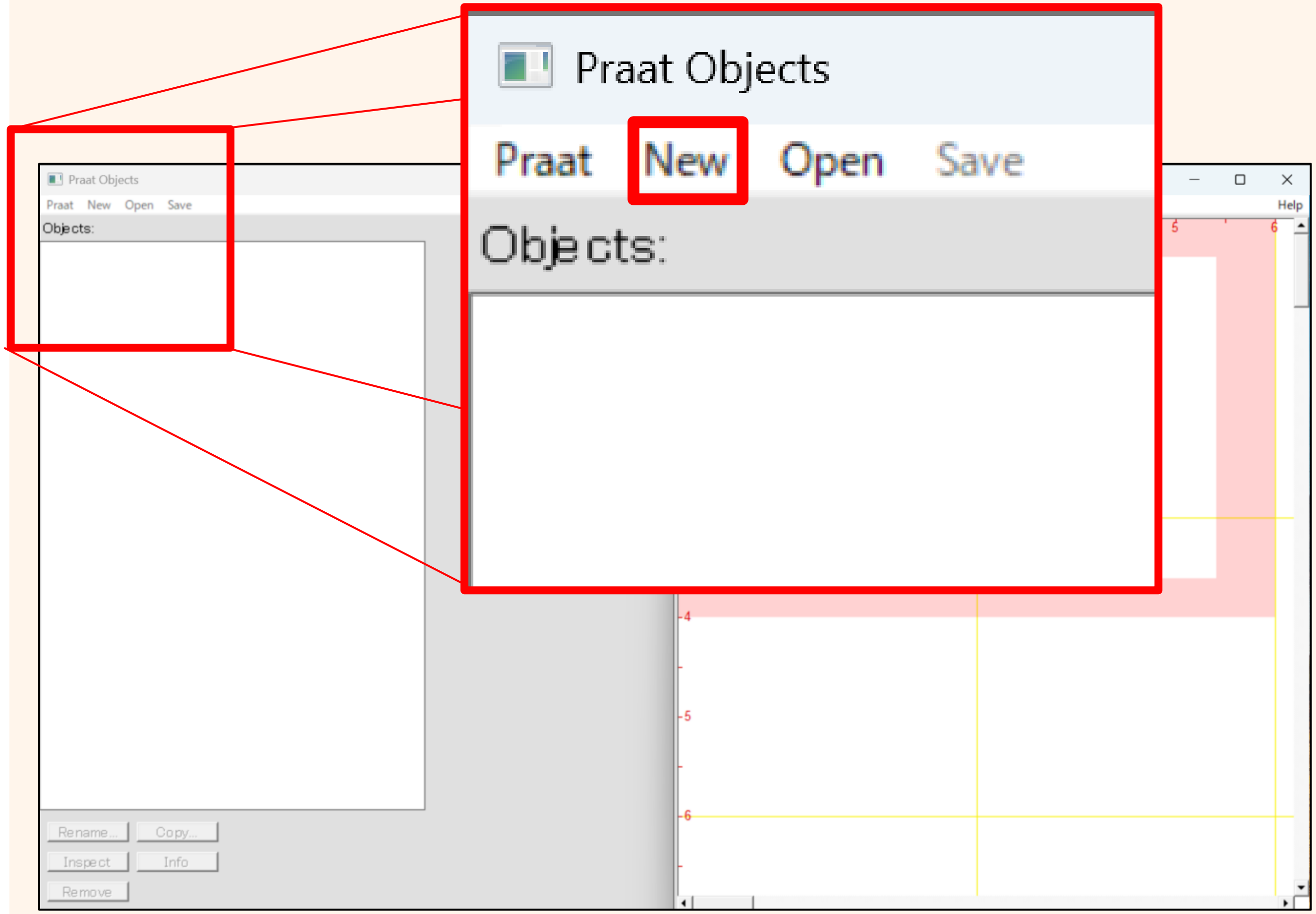
  → therefore: **allophones of <u>the same</u> underlying phoneme** in Spanish (representing /d/, the *idea* of a 'd')

- In English, /d/ and /ð/ are…
  - **contrastive** – making a difference in meaning between words
  - **unconditioned** – either can appear in the same phonetic context
  - **unpredictable** – you can't predict which version will be pronounced based solely on its position in a word

  → therefore: **allophones of <u>different</u> phonemes** in English

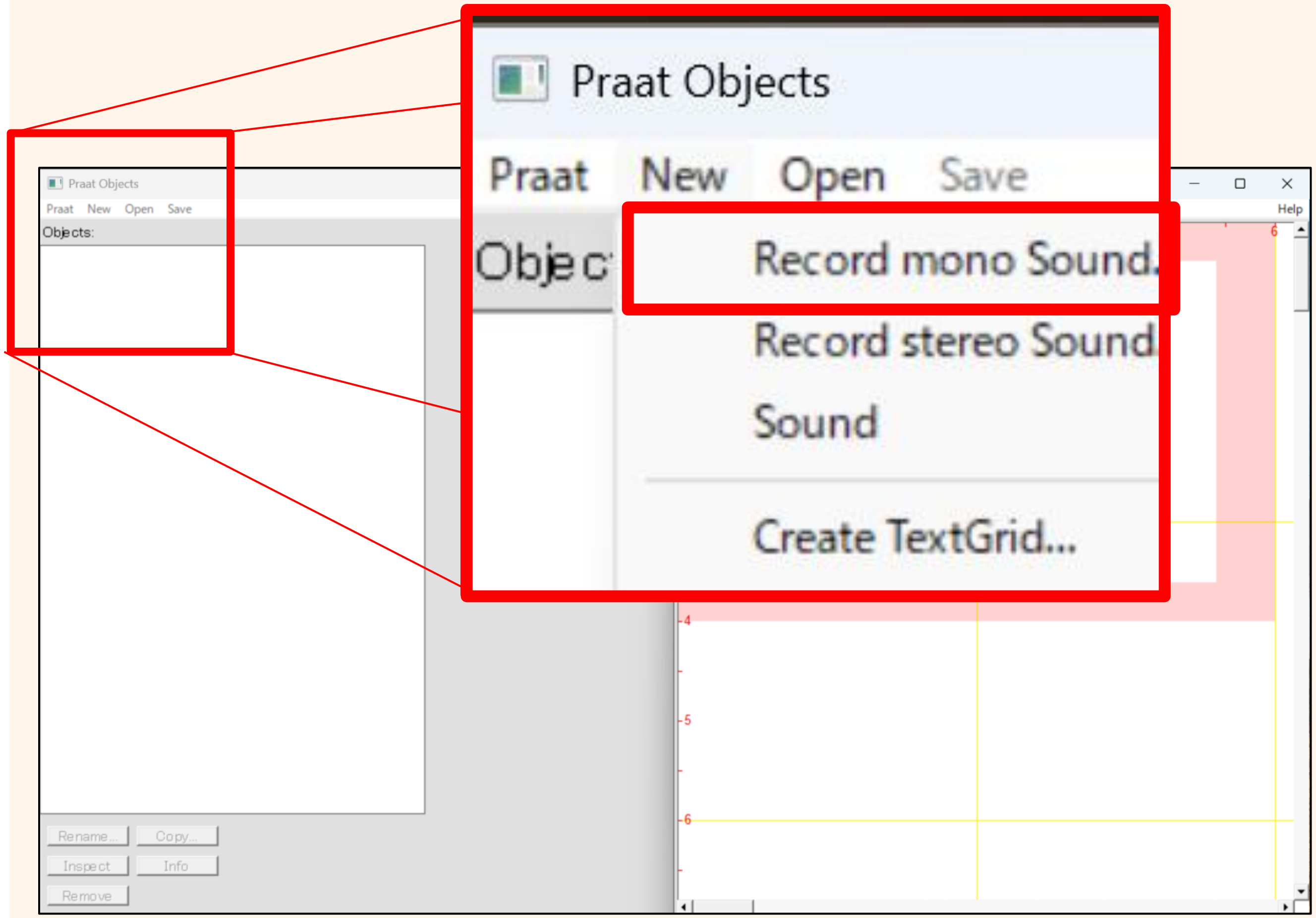
# Learning by Experiment

Step 1 Recording your voice



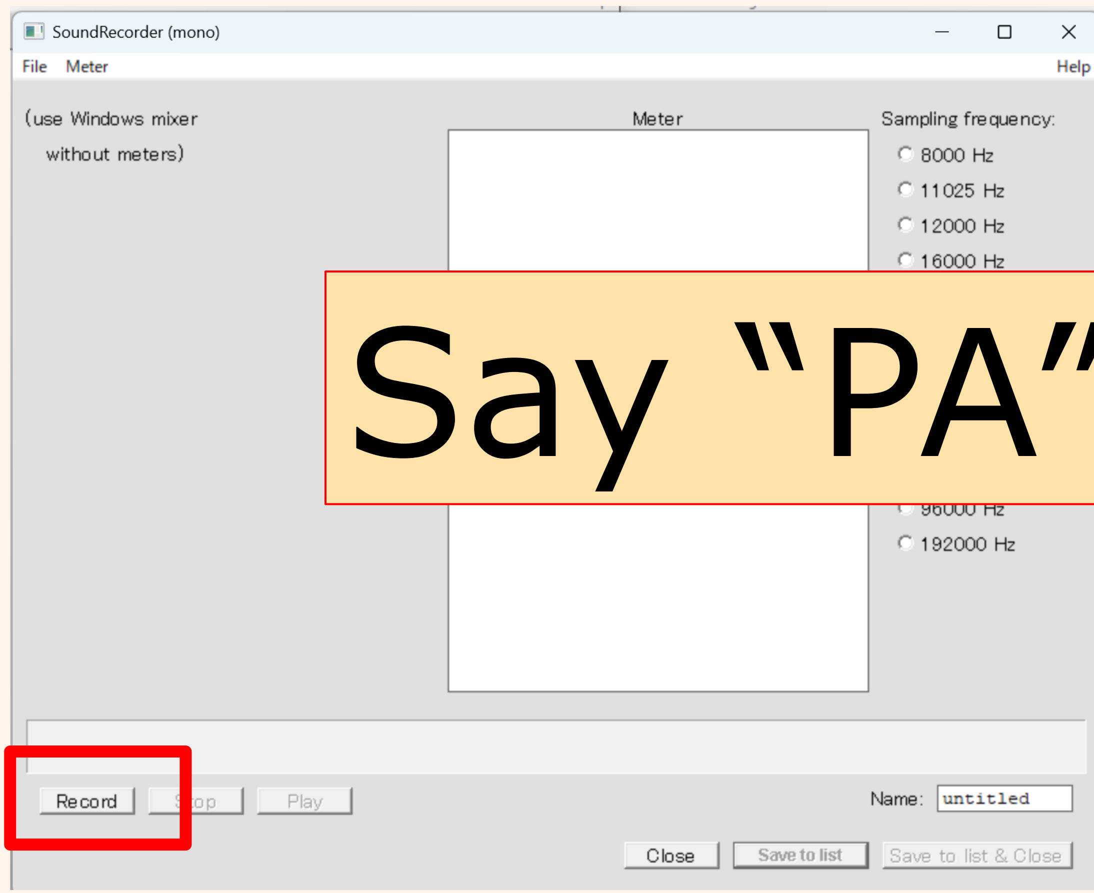
# Learning by Experiment

Step 1 Recording your voice



# Learning by Experiment

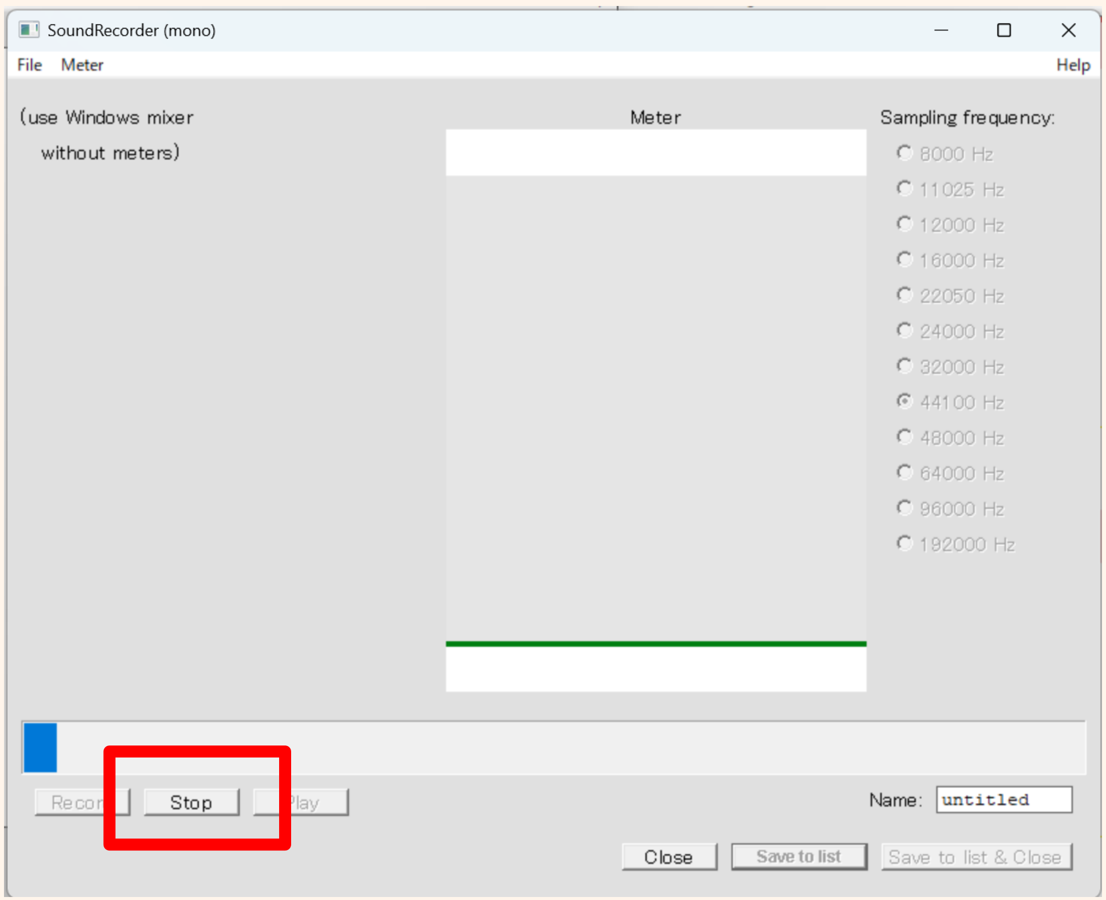
**Step 1** Recording your voice




Say “PA”!

# Learning by Experiment

**Step 1** Recording your voice



# Learning by Experiment

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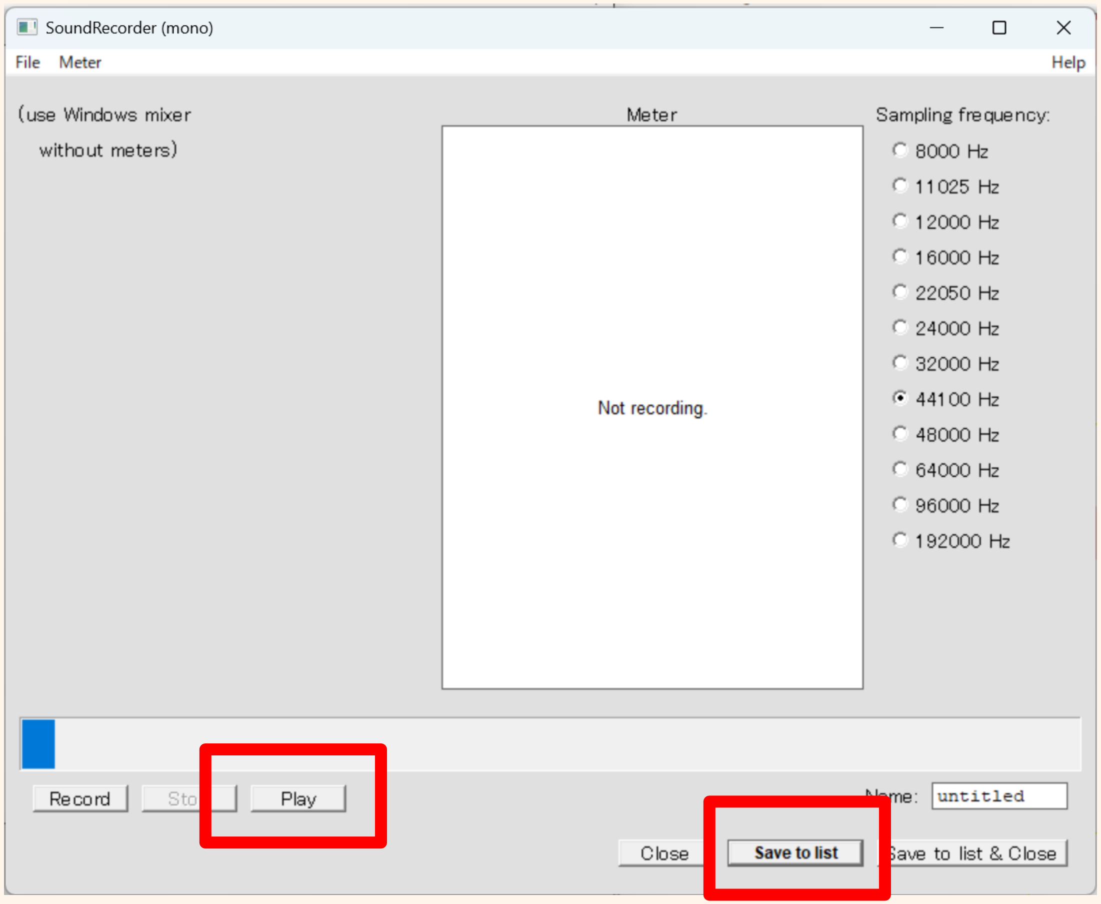
|---|---|



# Learning by Experiment

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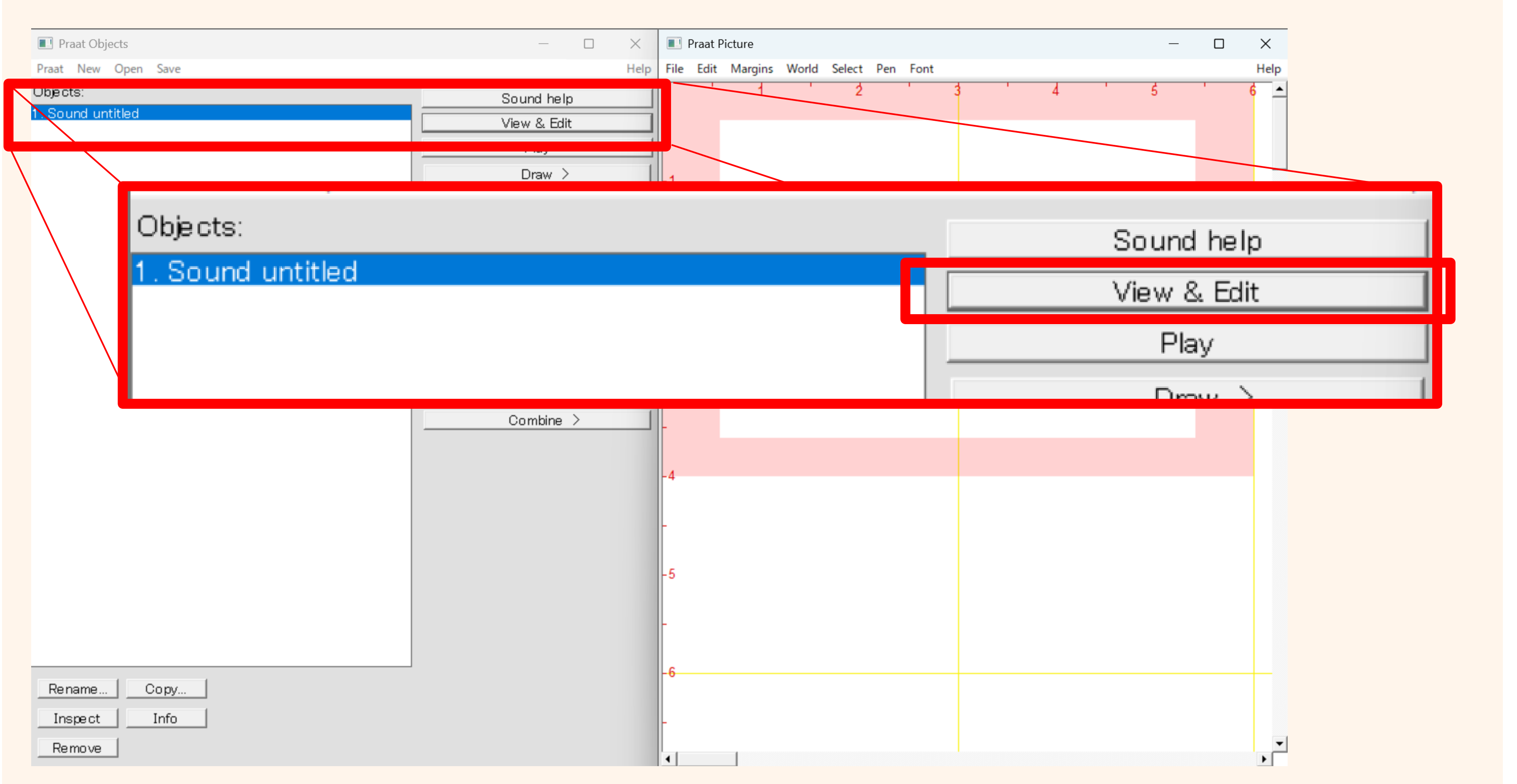
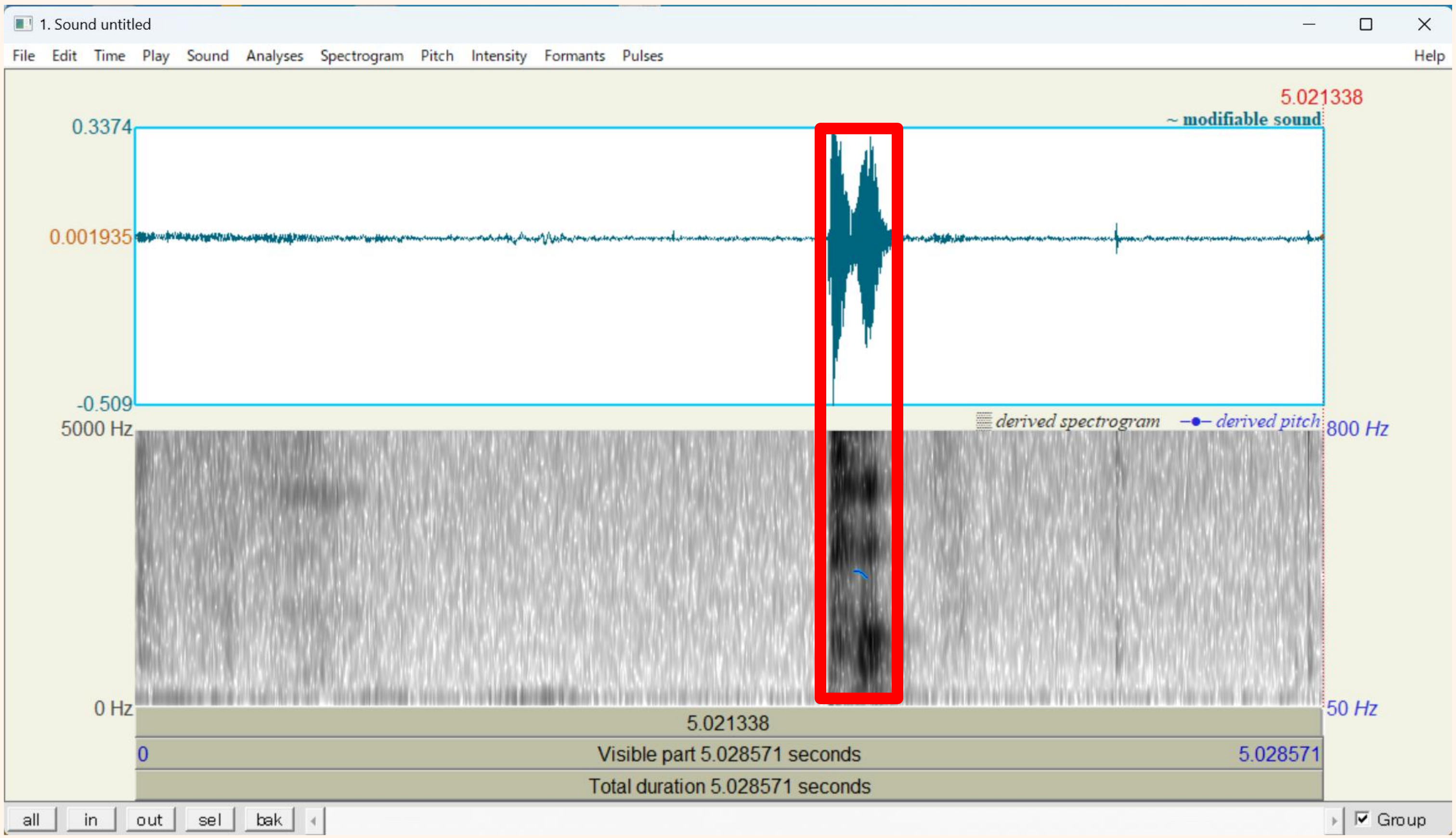
| --- | --- |



# Learning by Experiment

Step 2 Analyze your voice



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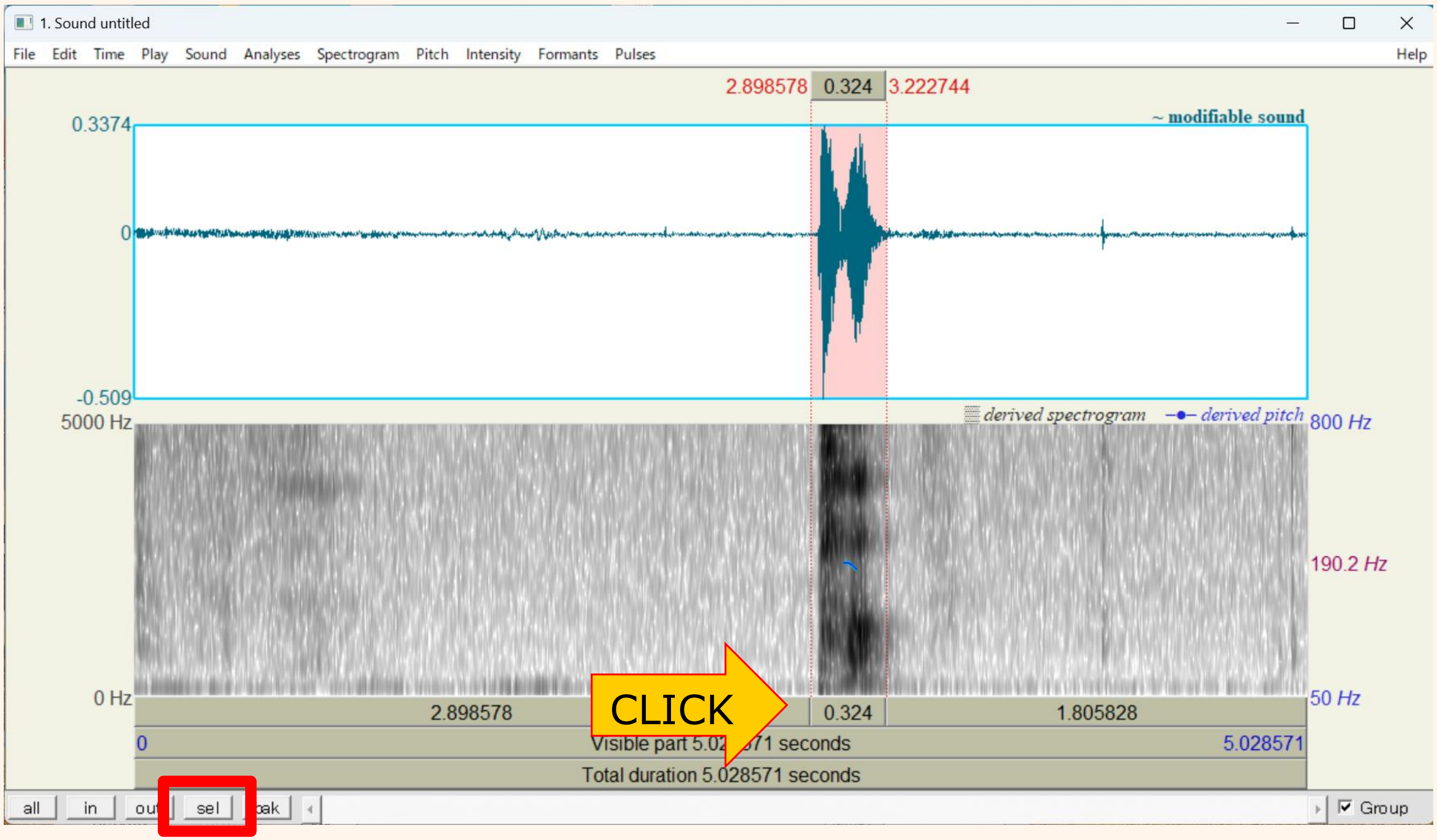

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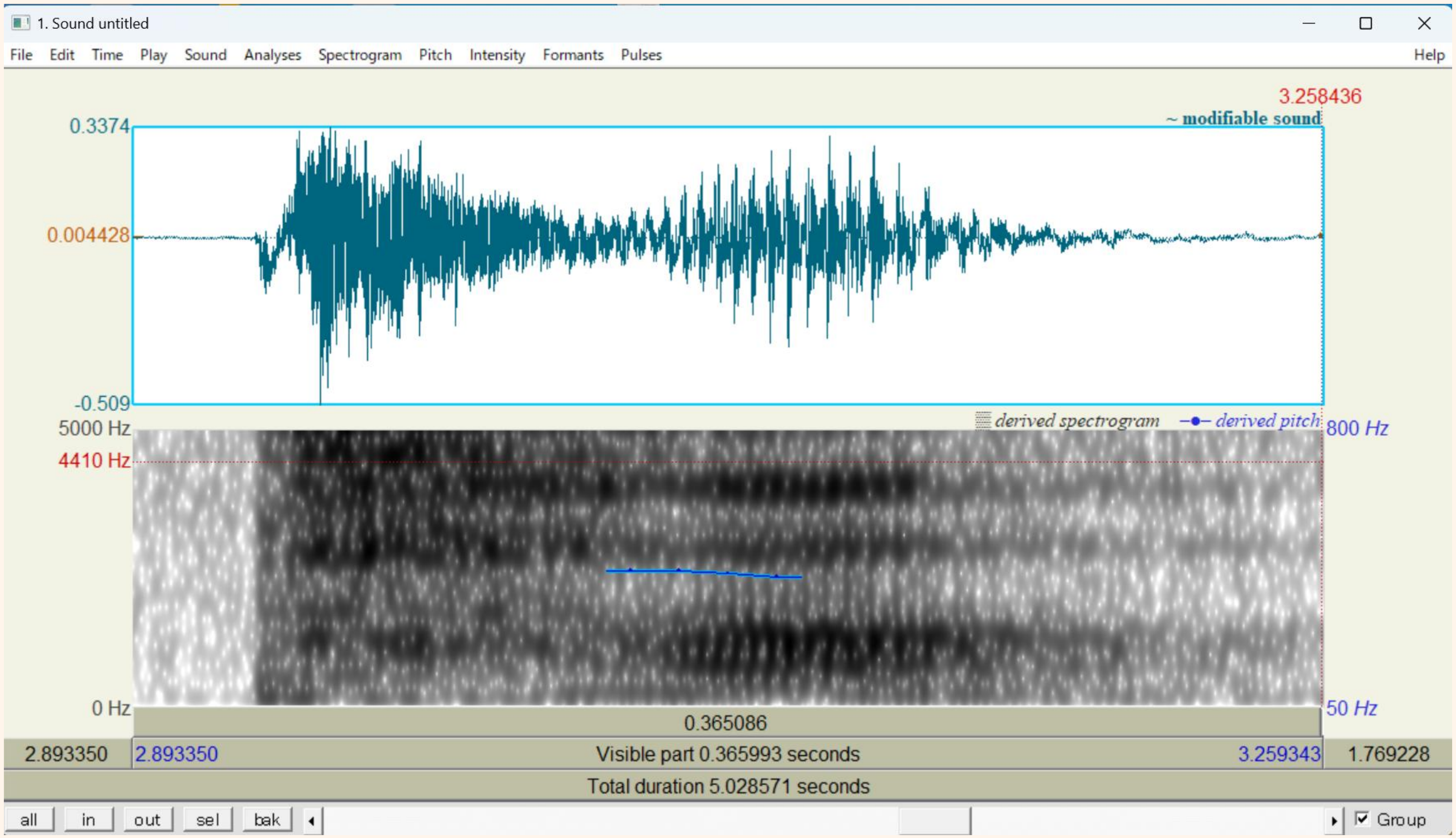
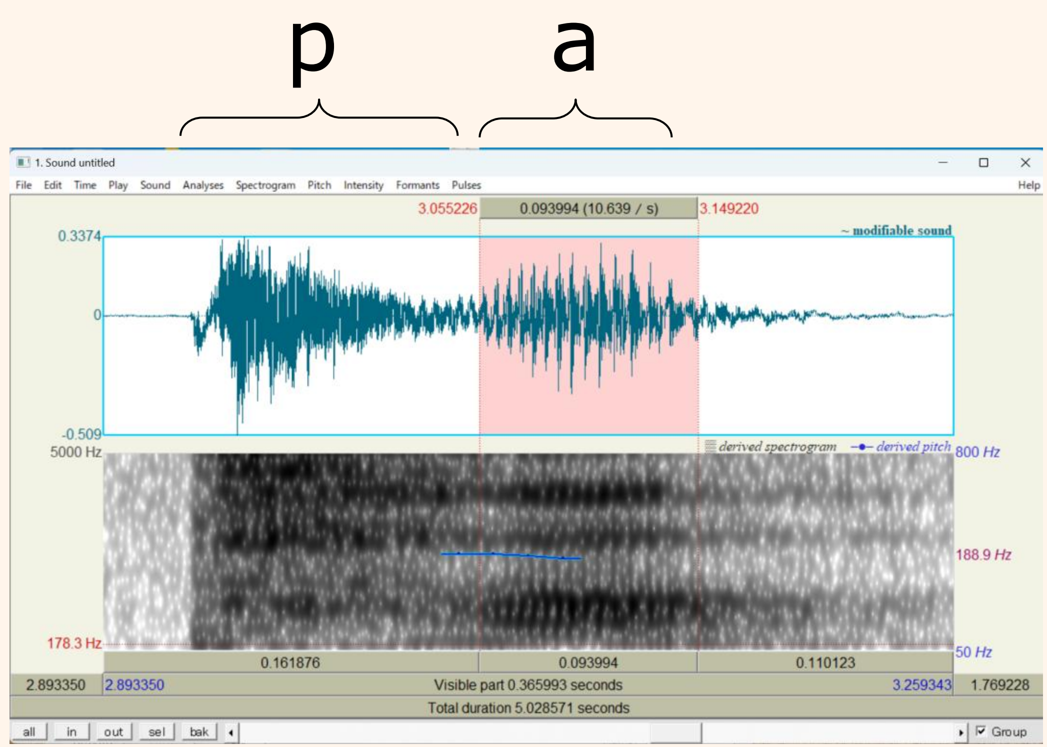


# Learning by Experiment

**Step 2** Analyze your voice

There must be a point where you stop hearing the [p] and it sounds just like a pure [a]. Let's find where that is.



# Learning by Experiment

**Step 3** Save your voice

There must be a point where you st
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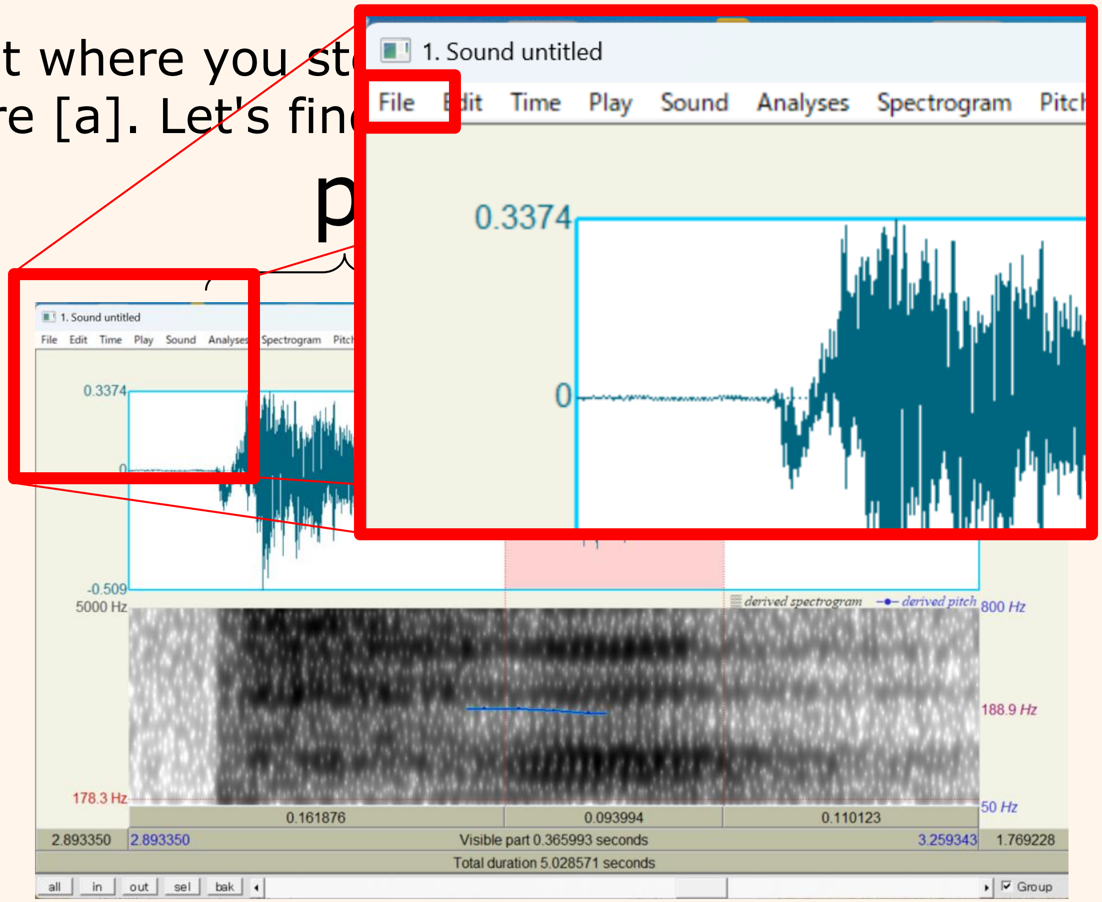

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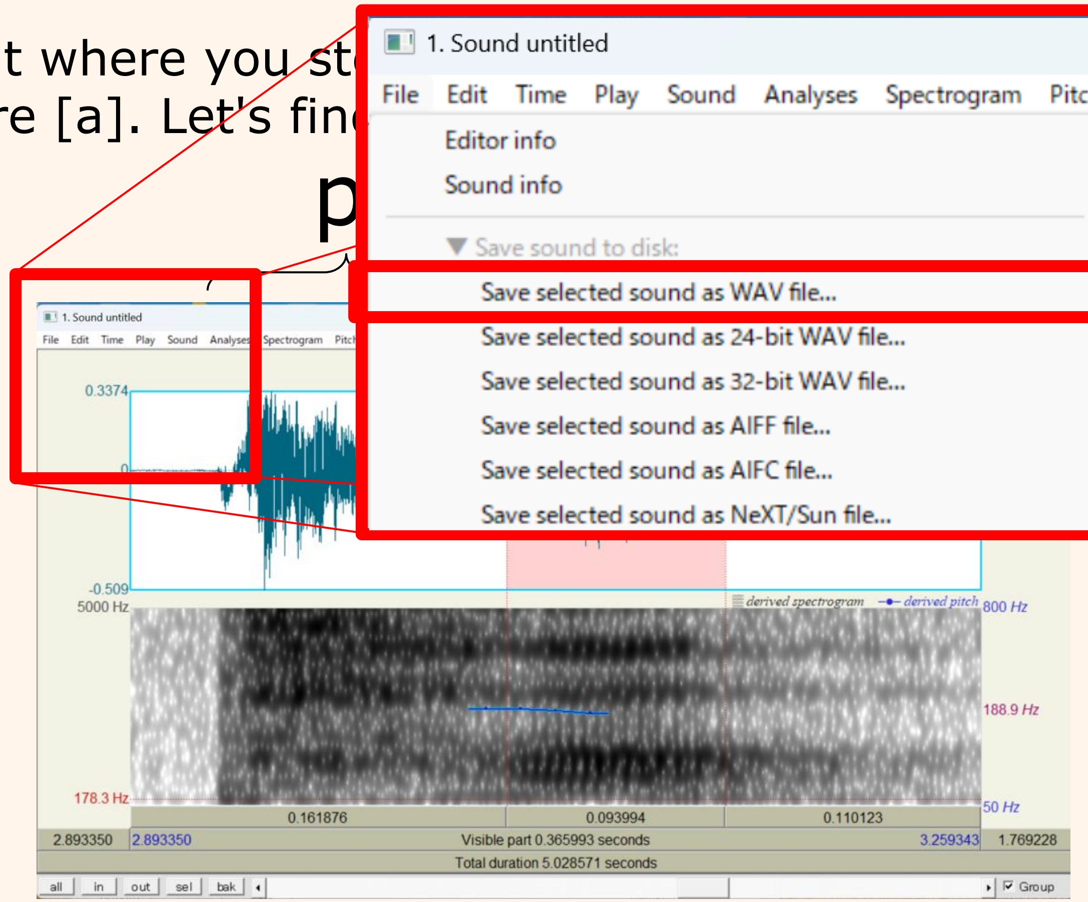

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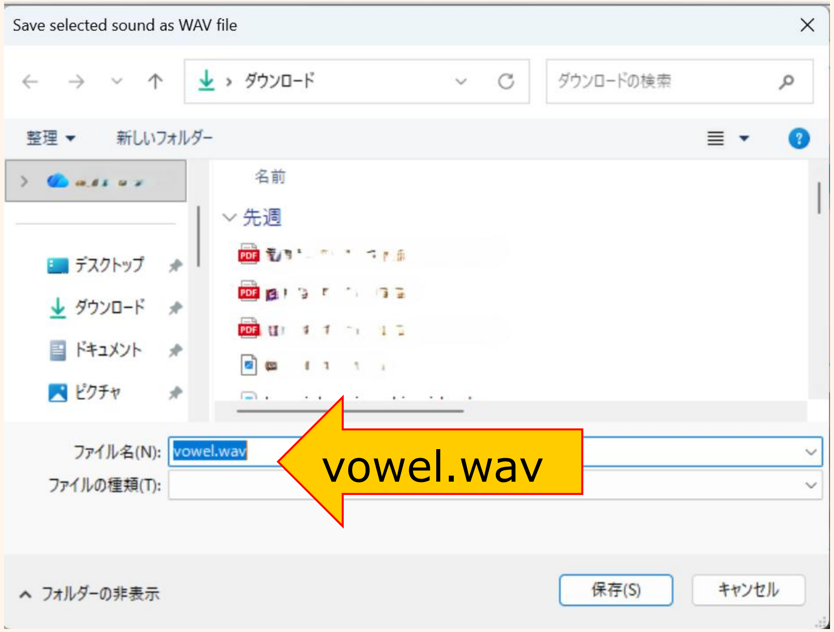
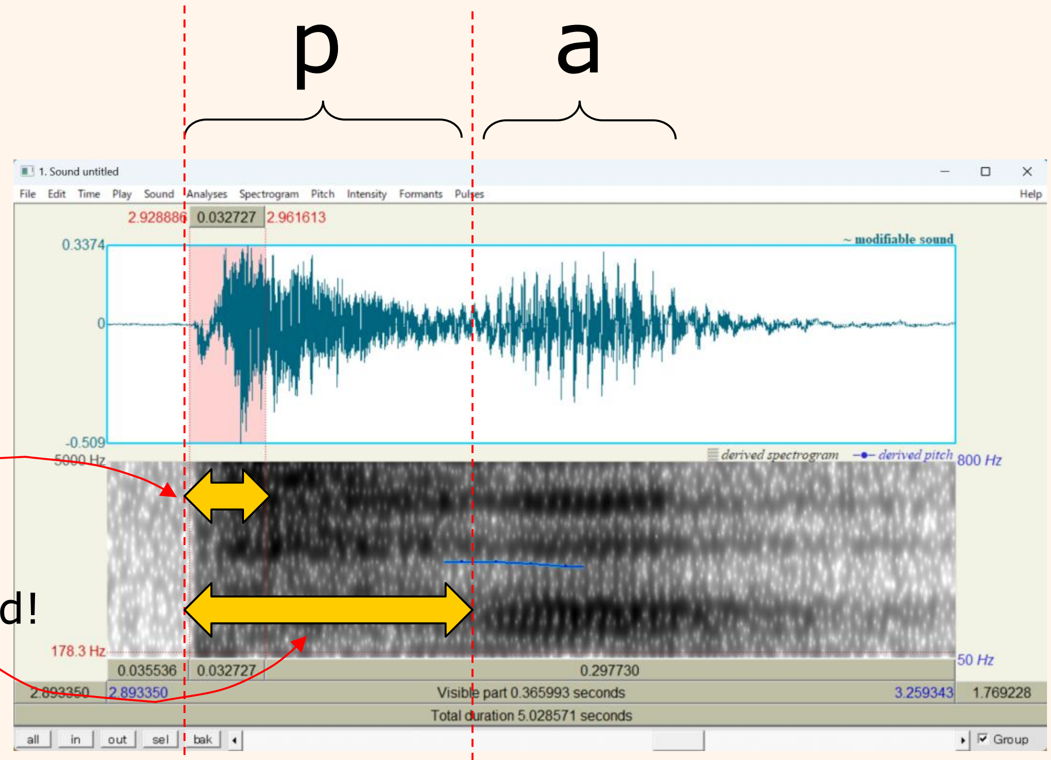


# Learning by Experiment

**Step 3** Save your voice

This time, we're going to save just a **part** of the [p]. Select only that **very first 'pop' or 'burst' sound**, and save it using the same method.




✓ Just this part!

* Not the whole 'P' sound!

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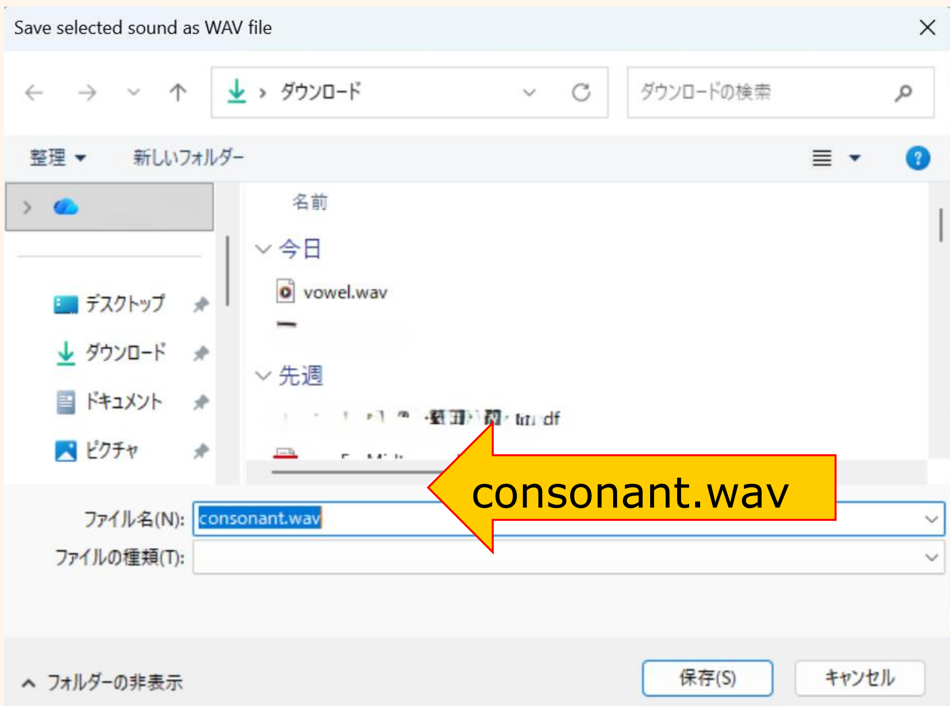
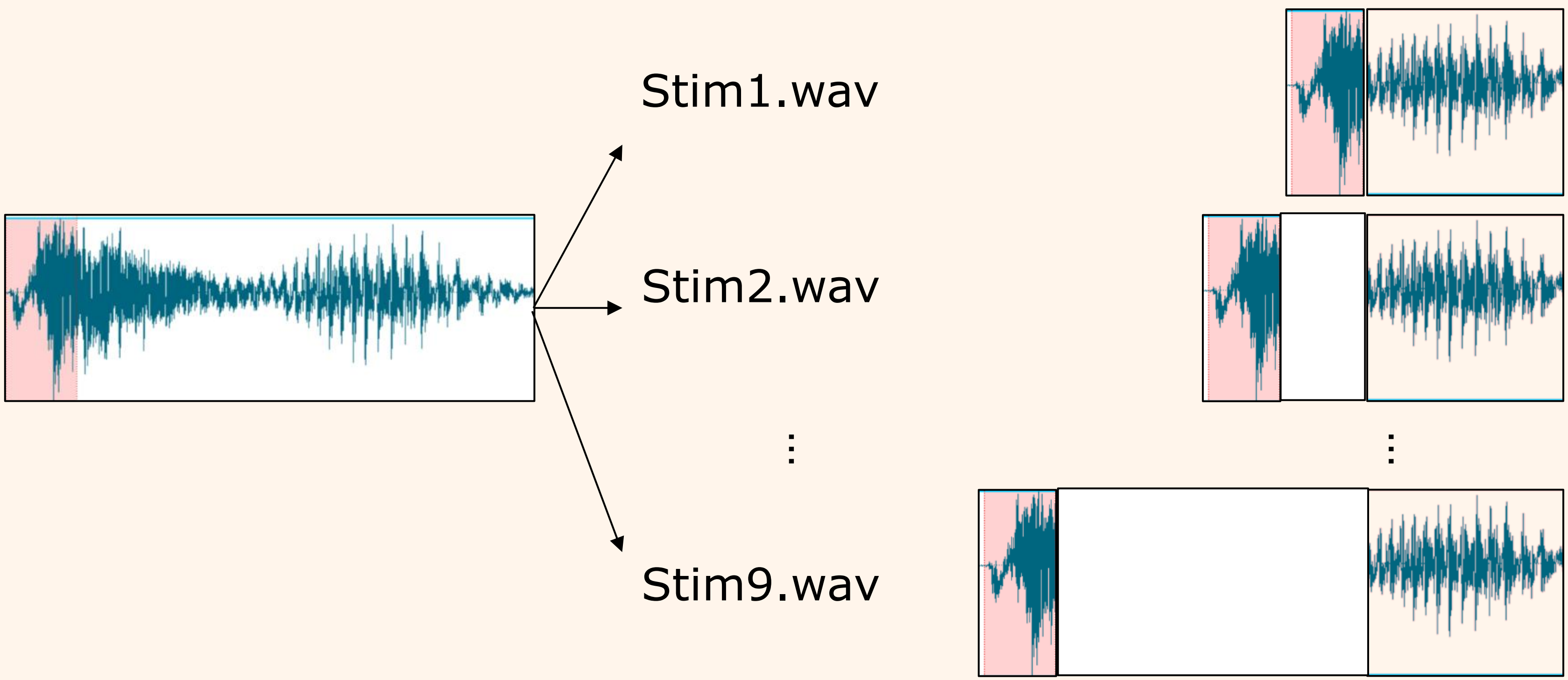


# Learning by Experiment

**Step 4** Creating stimuli

Now, we are going to create **nine synthetic sounds** by changing the duration between the burst and the onset of the vowel.



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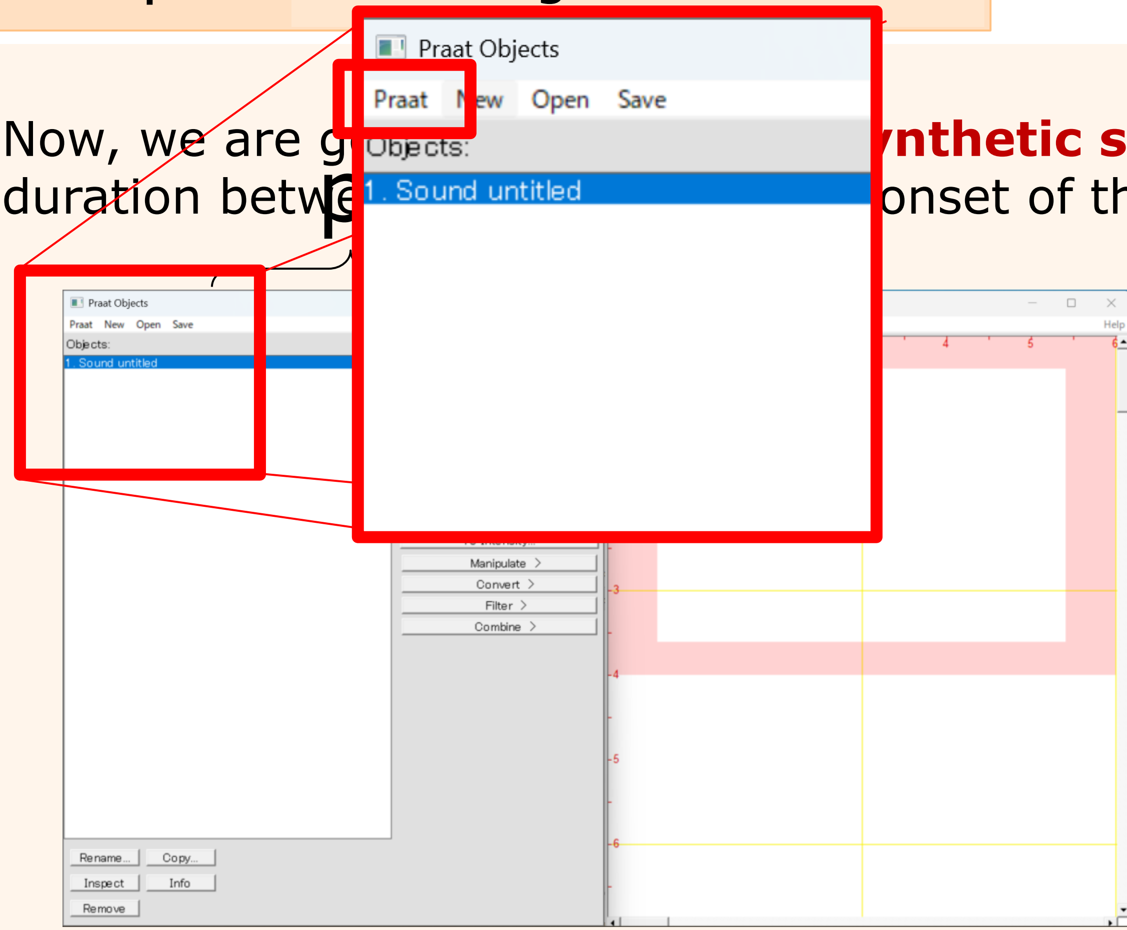

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**Step 4** Creating stimuli

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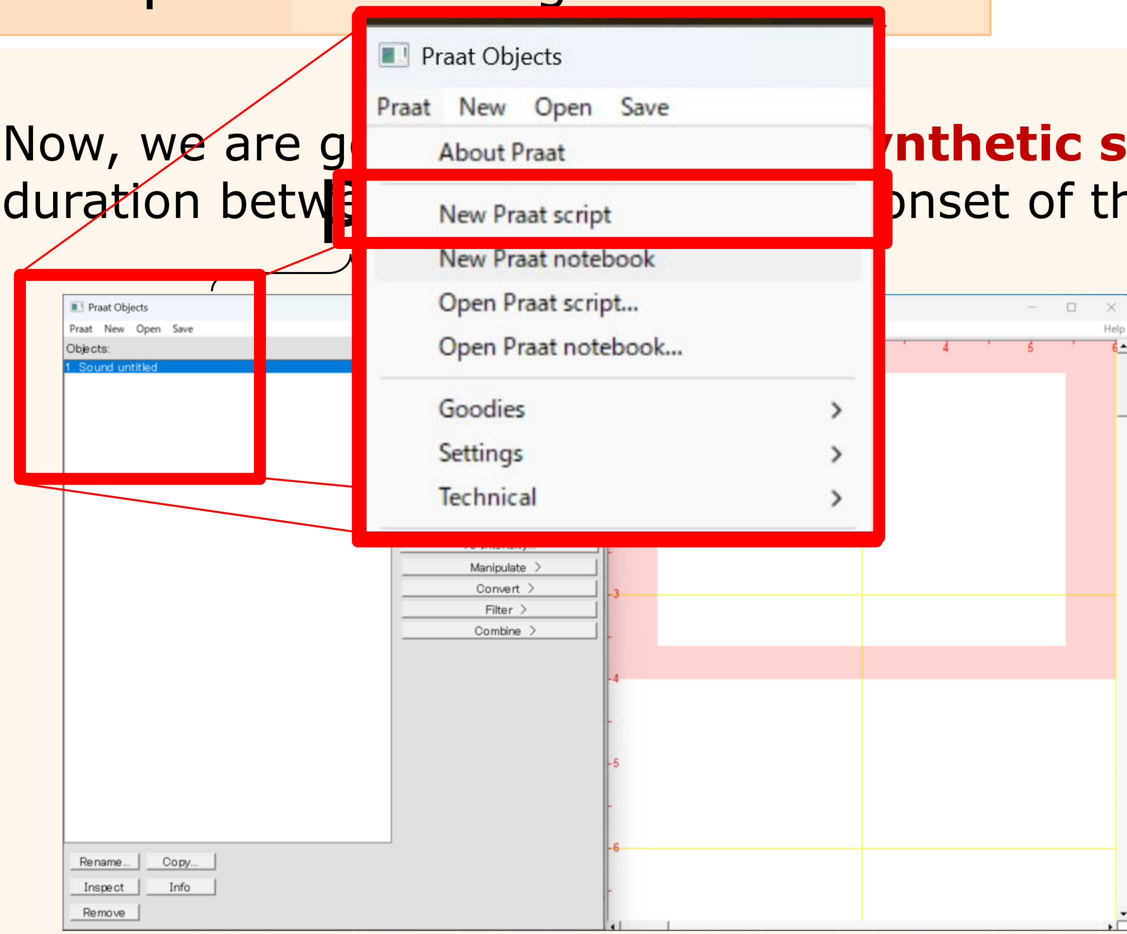

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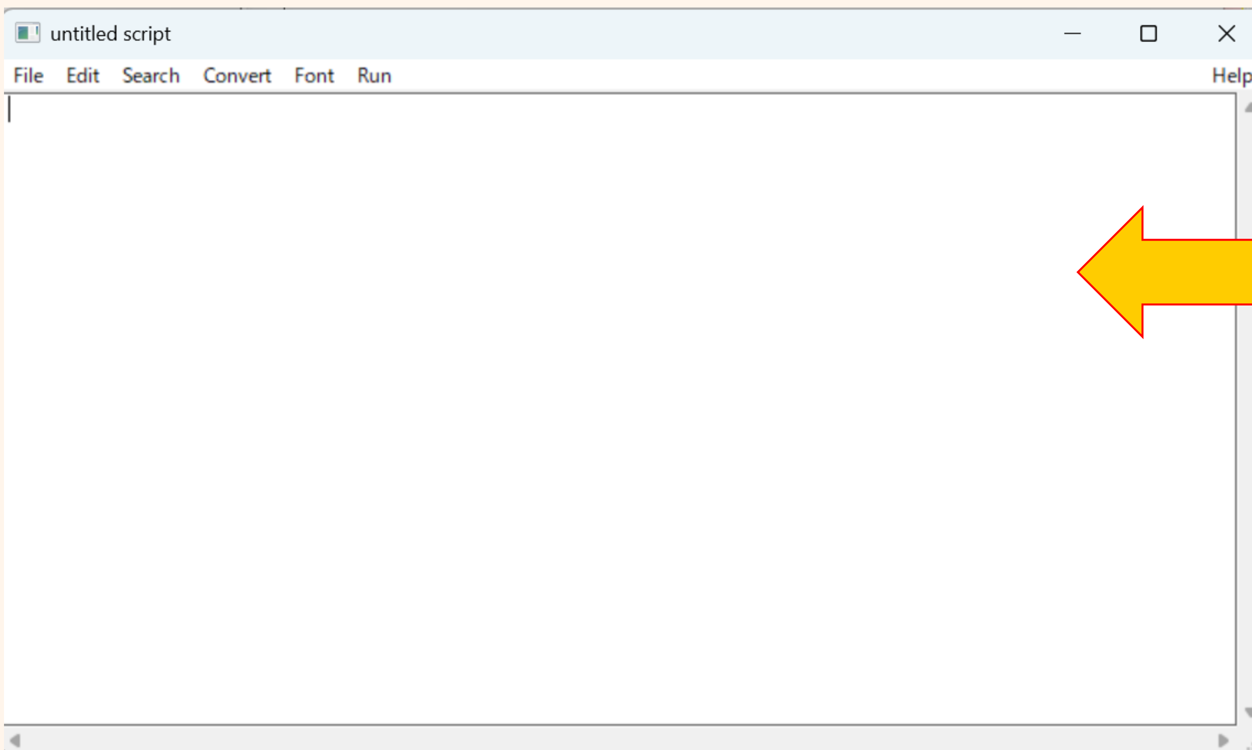



Please download **'Experiment.praat'** from the website and copy-paste the code.

# Learning by Experiment

**Step 4** Creating stimuli

Now, we are going to create **nine synthetic sounds** by changing the duration between the burst and the onset of the vowel.

```
# ============================================================================
# 1. SELECTION CHECK & INITIAL SETTINGS
# ============================================================================
# Check if exactly two Sound objects are selected in the object list.
# Object 1 (top) must be the 'burst' (the sharp release click part).
# Object 2 (bottom) must be the 'vowel' (the vocalic portion).
num = numberOfSelected("Sound")
if num <> 2
    exitScript: "Error: Please select exactly 2 Sound objects (1st: burst, 2nd: vowel)."
endif

# Store the IDs of the selected original sounds into variables for easy reference
burst_id = selected("Sound", 1)
vowel_id = selected("Sound", 2)

# Get the sampling frequency (Hz) of the burst to match it with the generated silence later
selectObject: burst_id
fs = Get sampling frequency

# ============================================================================
# 2. OUTPUT DESTINATION SELECTION
# ============================================================================
# Prompt the user to select the folder where the generated WAV files will be saved
```

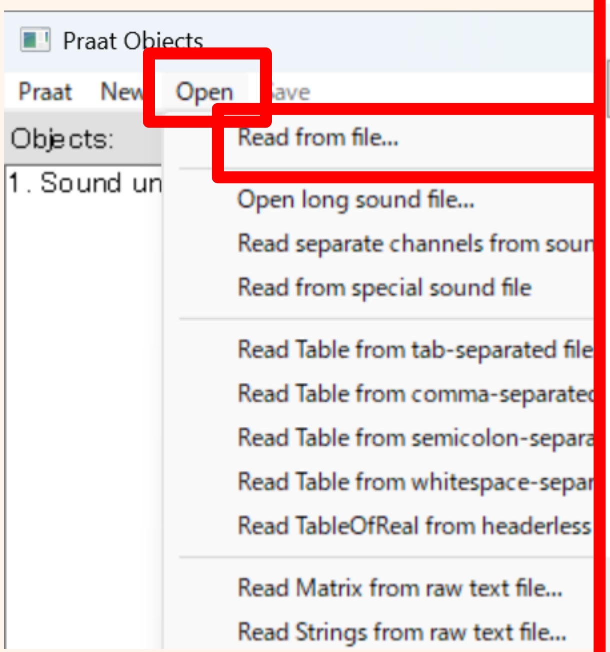
Please download **'Experiment.praat'** from the website and copy-paste the code.

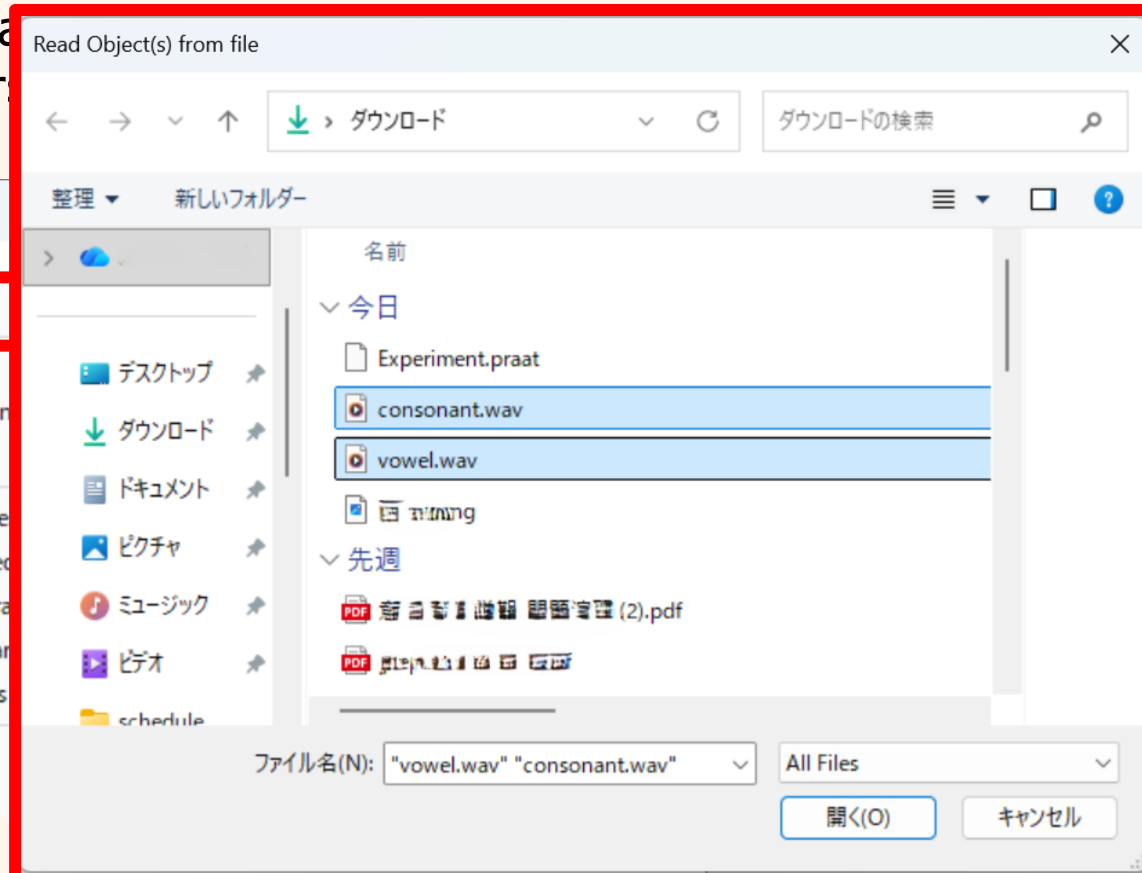
# Learning by Experiment

Select the files you created: "consonant.wav" and "vowel.wav".

## Step 4 Creating stimuli

Now, we are going to crea
duration between the bur



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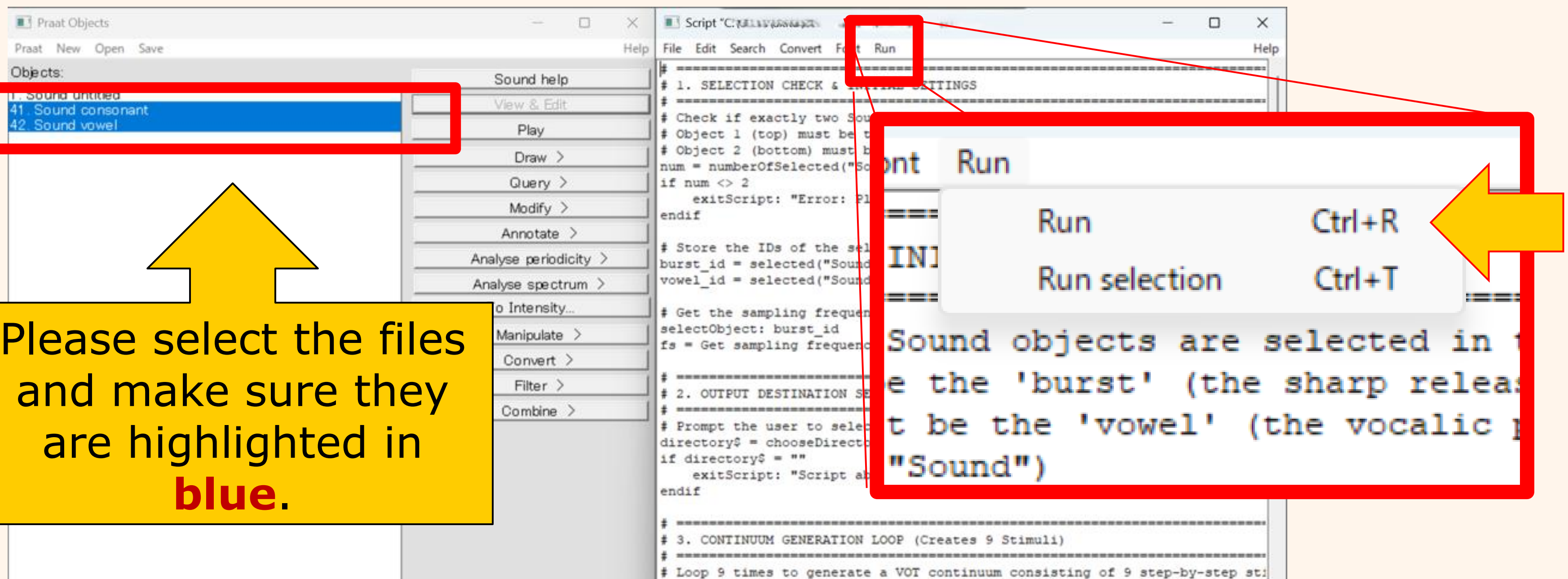


Please select the files and make sure they are highlighted in **blue**.

**Windows users:** *"Hold down the **Ctrl** key while clicking to select both files."*
**Mac users:** *"Hold down the **Command (⌘)** key while clicking to select both files."*

# Learning by Experiment

## Step 4 Creating stimuli

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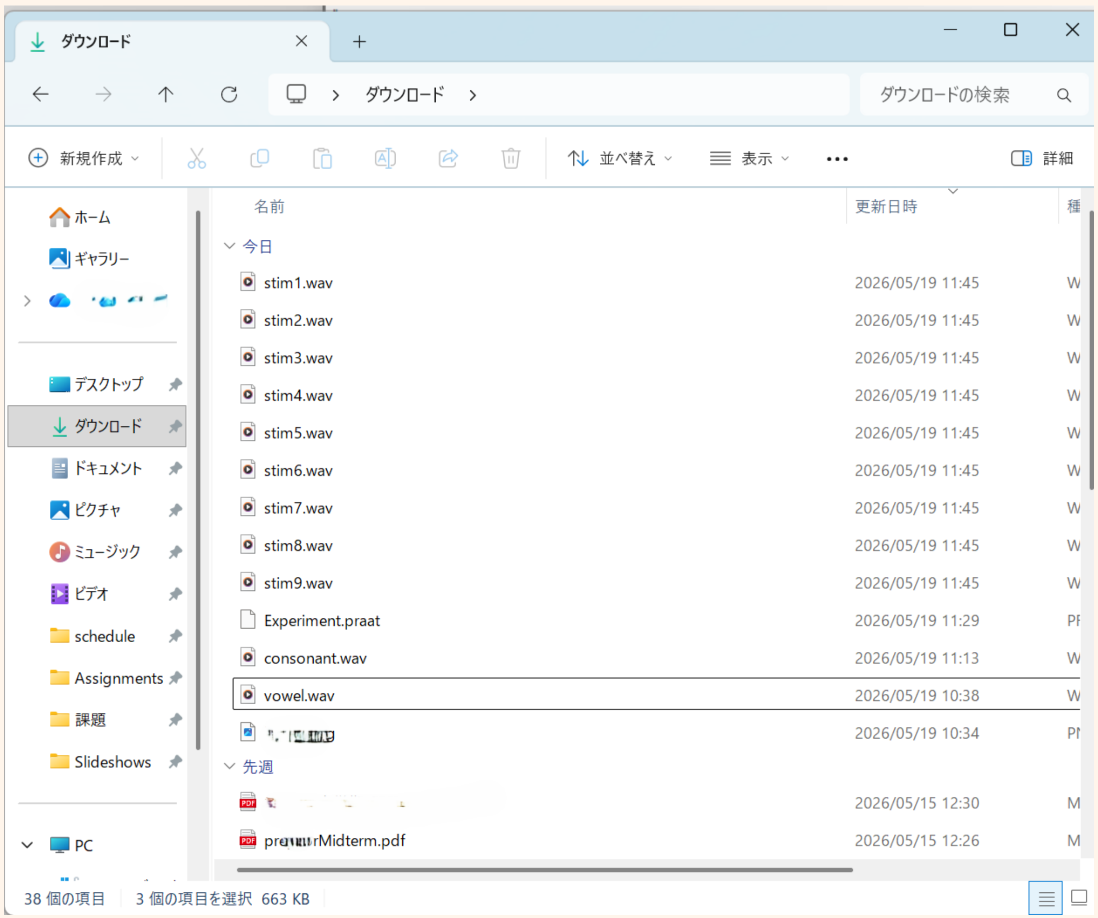


CONGRATS!

# Learning by Experiment

Select the files you created: "consonant.wav" and "vowel.wav".

## Step 5 Distributing your exp

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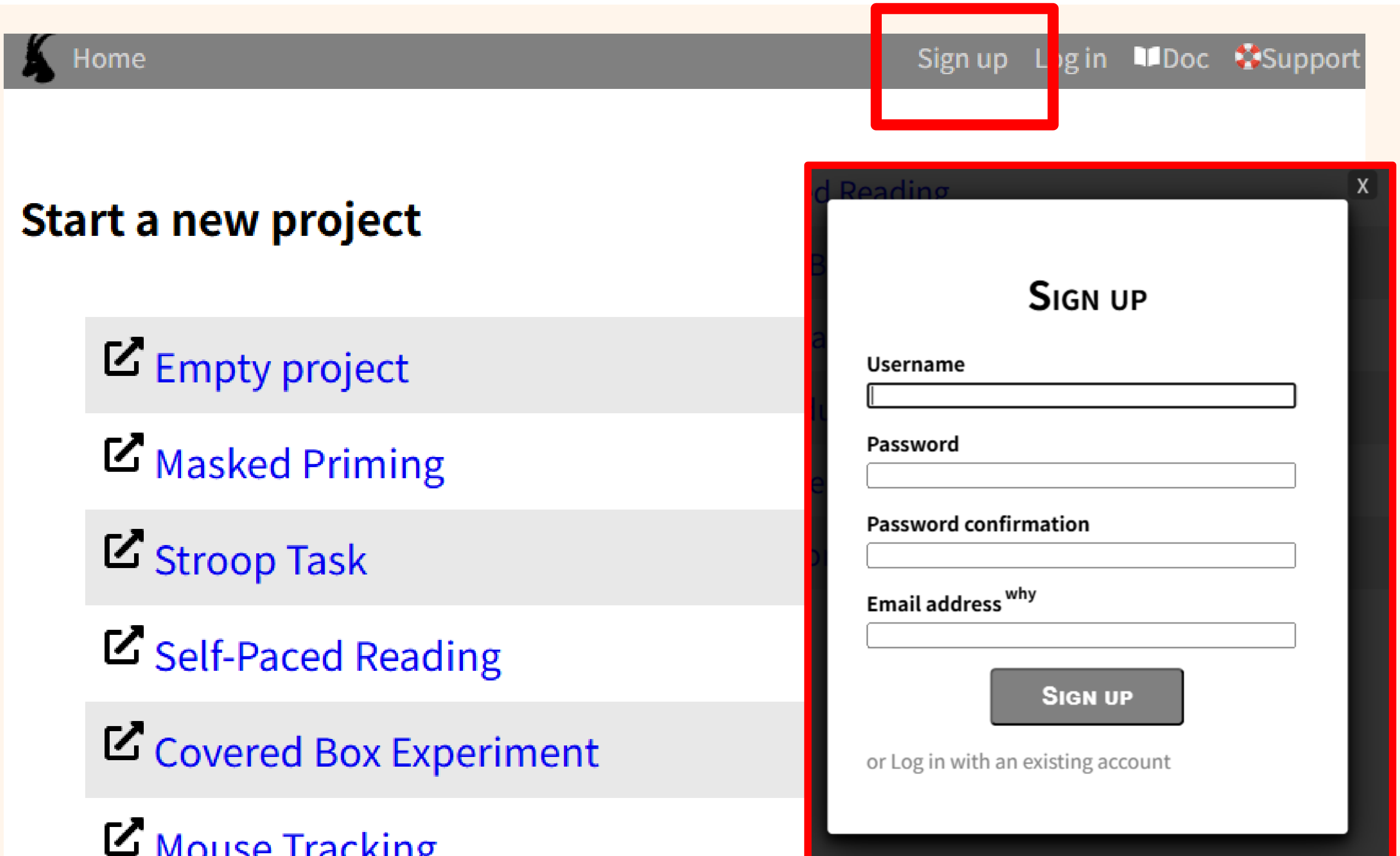
Step 5 Distributing your exp



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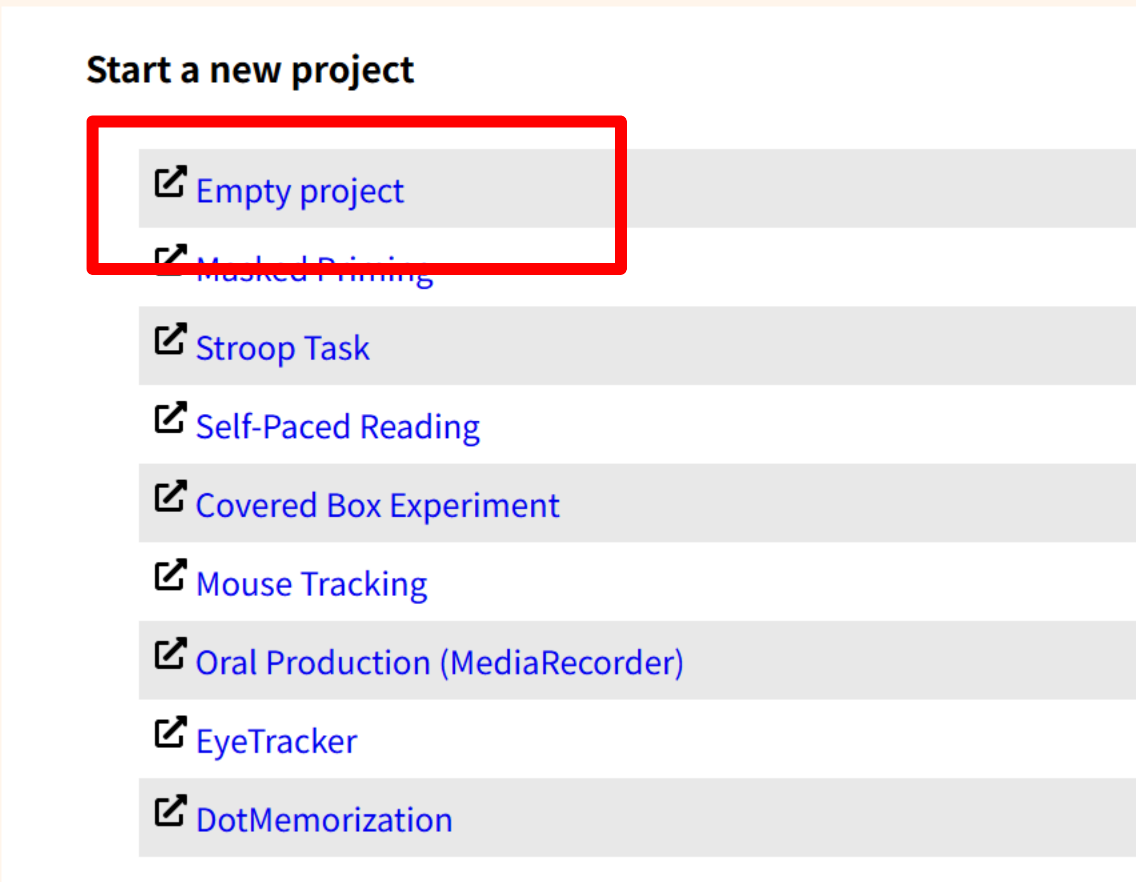

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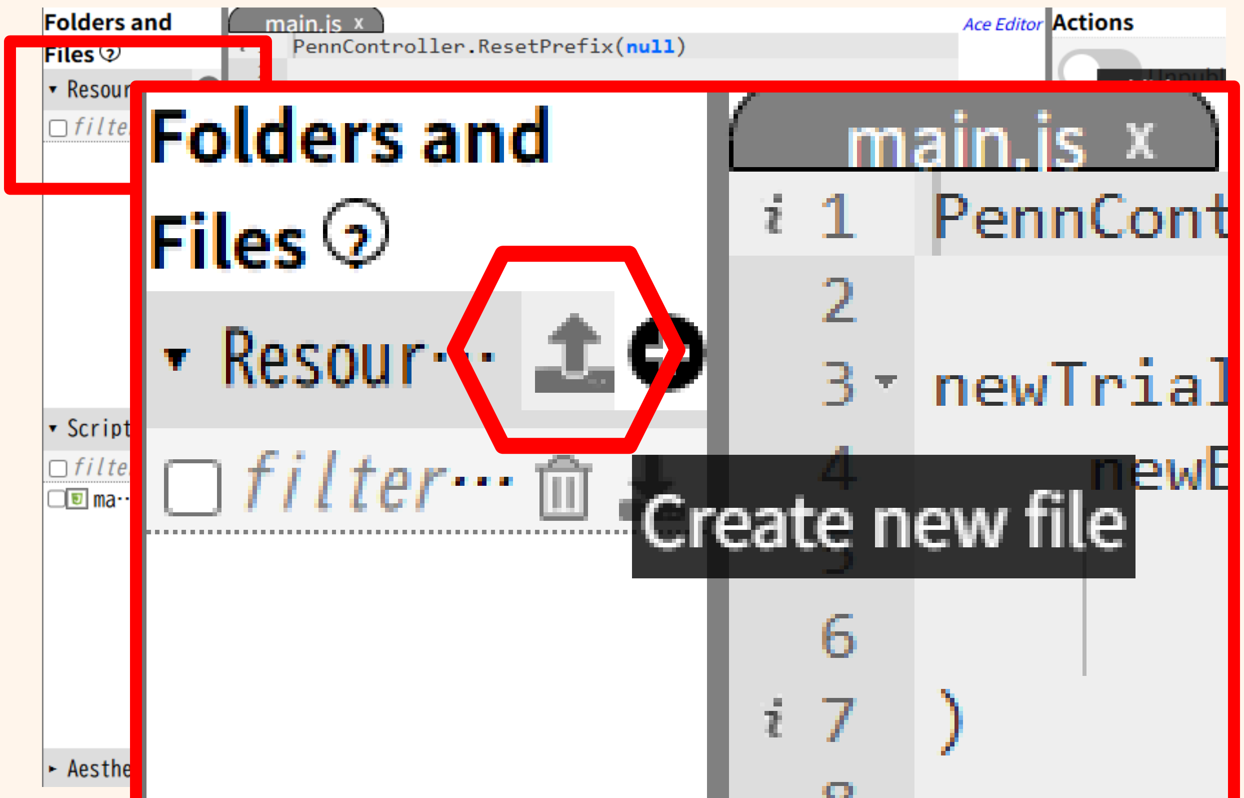
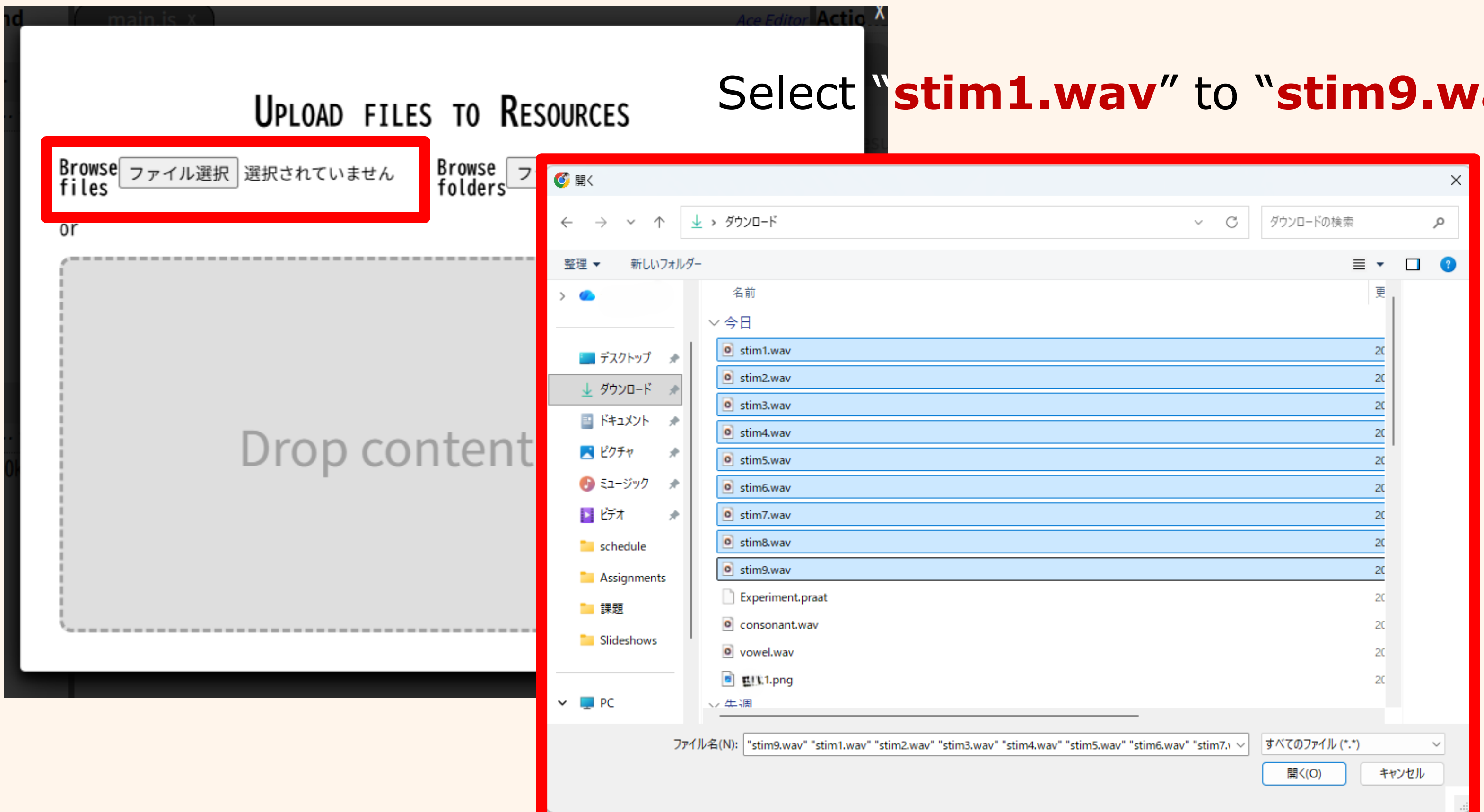


# Learning by Experiment

## Step 5 Distributing your exp

Select the files you created: “consonant.wav” and “vowel.wav”.

Select “**stim1.wav**” to “**stim9.wav**”



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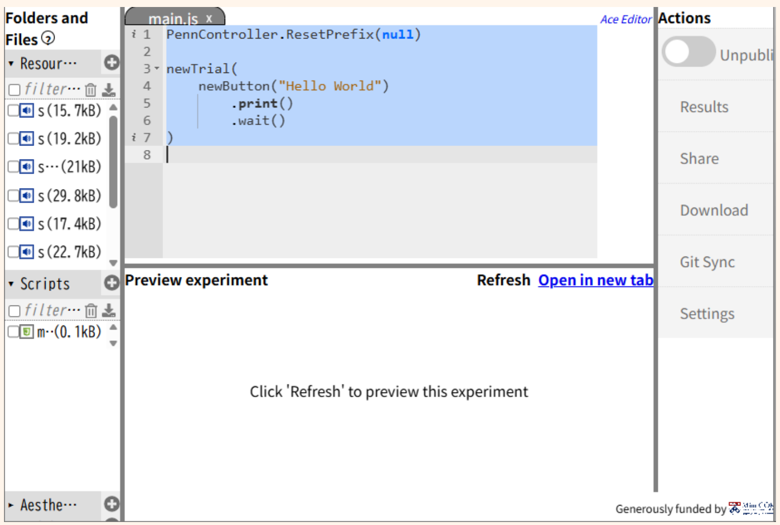



Delete the code in the main panel, and then, …

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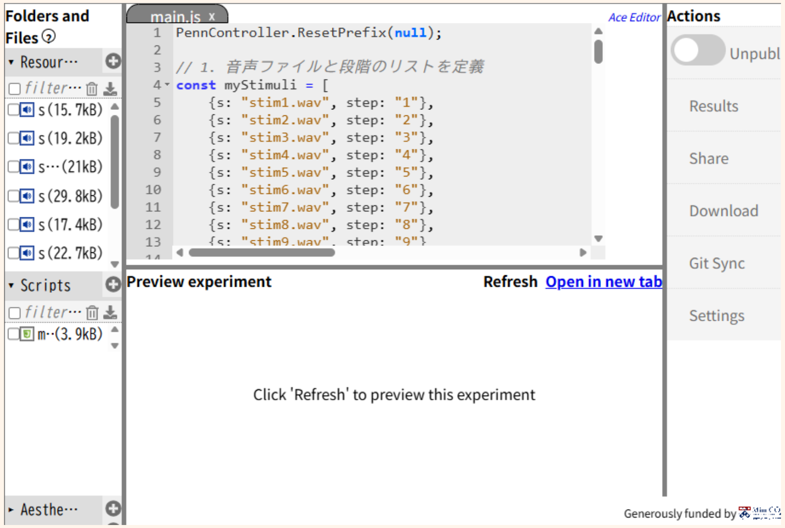
## Step 5 Distributing your exp


Delete the code in the main panel, and then, …

download ‘**PCIbex.txt**’ from the website and copy-paste the code.

# Learning by Experiment

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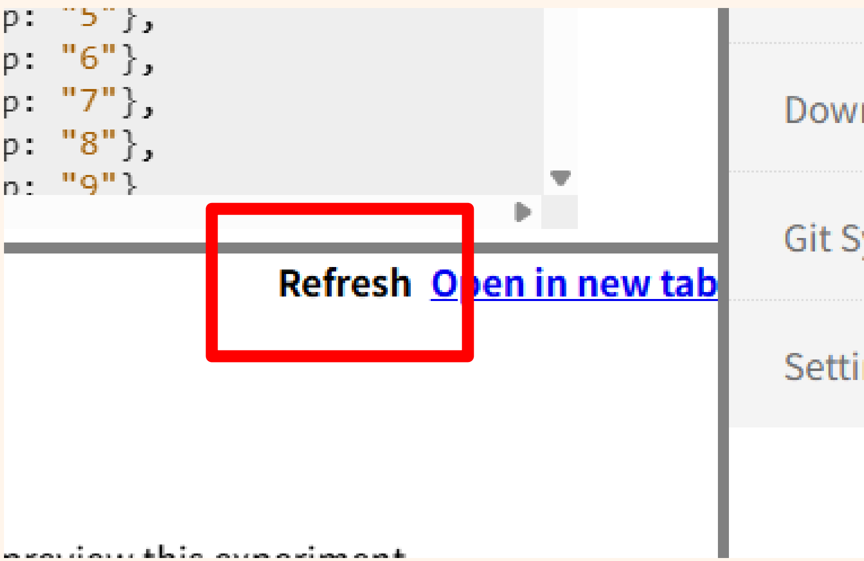
**Step 5** Distributing your exp




Click "**Refresh**" to check if it's ready for distribution.

# Learning by Experiment

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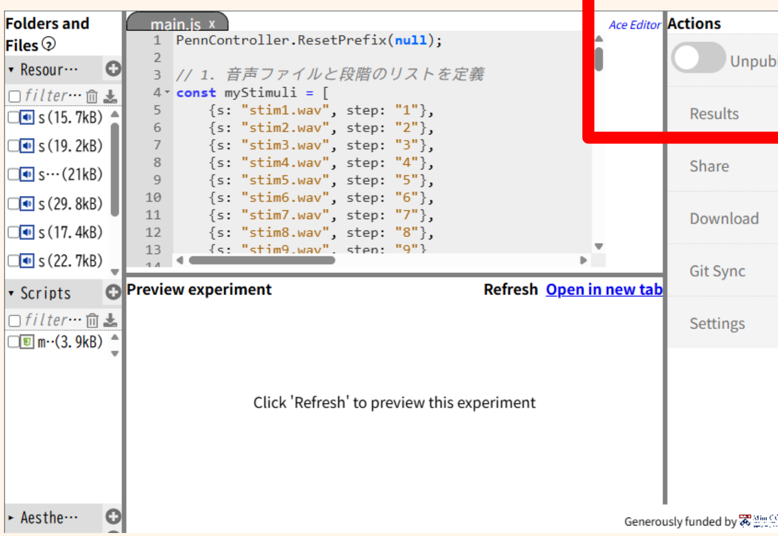


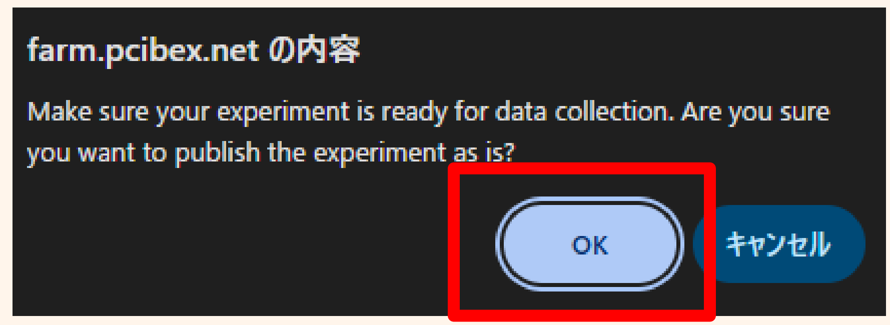

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